test

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June 24, 2026

fundamental theorem of calculus, part ii let $f(x)$ be a continuous function on $[a, b]$. then $A(x) = \int_a^x f(t)\,dt = f(x)$, that is, $A'(x) = f(x)$, or equivalently, $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$ furthermore, $A(x)$ satisfies the initial condition $A(a) = 0$

arithmetic: $\sum_{k=1}^{n}(a + (k-1)d)$

1. $S_n = a + (a + d) + (a + 2d) + \ldots + (a + (n-1)d)$
2. $S_n = (a + (n-1)d) + (a + (n-2)d) + \ldots + a$
3. $2S_n = [a + (a + (n-1)d] + [(a + d) + (a + (n-2)d] + \ldots$
4. $2S_n = n(2a + (n-1)d)$
5. $S_n = \frac{n}{2}(2a + (n-1)d)$

geometric: $\sum_{k=1}^{n} ar^{k-1}$

1. $(S_n - rS_n) = (a + ar + ar^2 + \cdots + ar^{n-1}) - (ar + ar^2 + ar^3 + \ldots + ar^n)$
2. $S_n - rS_n = a - a^r$
3. $S_n(1 - r) = a - ar^n$
4. $S_n = \frac{a-ar^n}{1-r} = \frac{a(1-r^n)}{1-r}$
   $r \neq 1$

power series: $\sum_{n=0}^{\infty} a_n(x - c)^n$ every polynomial of degree d can be expressed as a power series around any center c, where all terms of degree higher than d have a coefficient of zero.

- centered around c=0 $f(x) = x^2 + 2x + 3 \Rightarrow f(x) = 6 + 4(x-1) + 1(x-1)^2 + 0(x-1)^3 + \ldots$
- the geometric series formula $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \ldots$ valid for $|x| < 1$